Chapter 10: Rotational Motion

The following subjects will be discussed

- Rotational/angular Motion Variables.
- Equations of Motion.
- Inertia of an Object.
- Kinetic Energy.
- Torque and Newton's Laws.
- Work in Angular Motion.

Newton's Second Law for Rotation

Here, an external force is applied to cause rotation. However, only the tangential component of the applied force would cause rotation.

$$F_t = F \sin\theta$$

We need to relate this force to the rotational motion variables.

$$F_t = m\, a_t$$

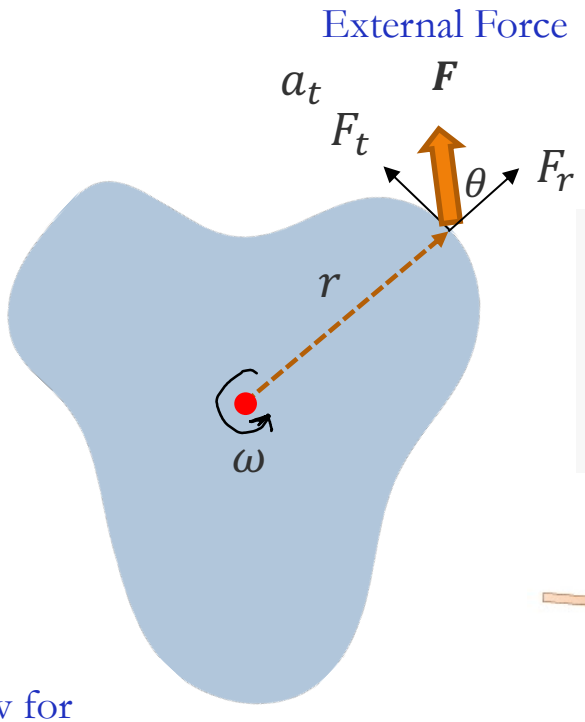
$$F_t = m\, \alpha\, r$$

$$F_t\, r = (mr^2)\, \alpha$$

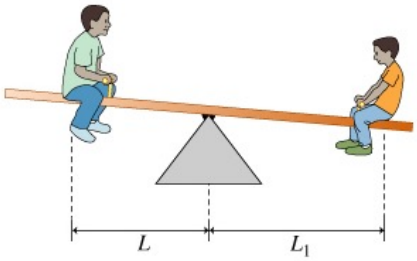
We define "Torque" → $\tau = I\, \alpha$

Units N.m

Newton's Second Law for Rotational motion







$$\tau_{net} = F_{t_{net}}\, r$$

Positive torque, counter-clockwise (CCW) motion.

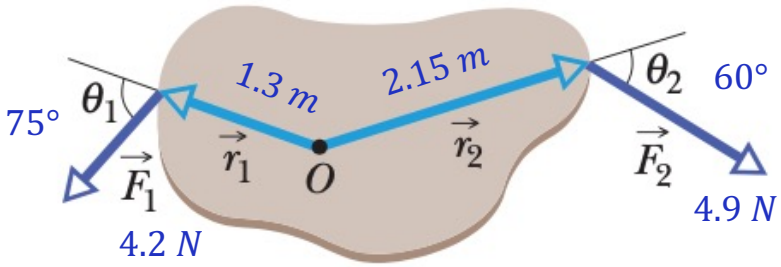
Negative torque, clockwise (CW) motion.

Examples

1) In the set-up below, find the net torque around point O.

$$\tau_1 = F_1 \sin\theta_1 \, r_1$$

$$\tau_1 = 4.2 \sin 75 \, (1.3) = 5.27 \; N.m \;\; (CCW)$$



$$\tau_2 = F_2 \sin\theta_2 \, r_2$$

$$\tau_1 = 4.9 \sin 60 \, (2.15) = 9.12 \; N.m \;\; (CW)$$

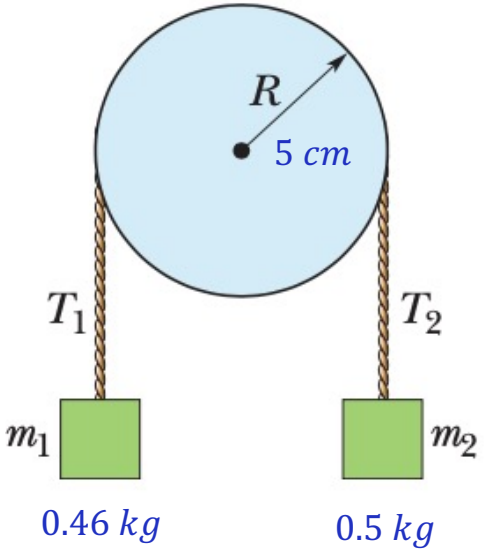
$$\tau_{net} = 5.27 - 9.12 = -3.85 \; N.m \; (CW)$$

Examples

2) The system below is initially at rest. At t=0, mass 2 falls 75 cm in 5 s, without the rope slipping (pulley rotates).

Find the inertia of the pulley.

We can find the inertia from Newton's law $\tau = I\,\alpha$, but we need the angular acceleration and torque.



a) Find linear acceleration

$$y - y_o = v_o t + \frac{1}{2}at^2 \quad \rightarrow \quad 0.75 = \frac{1}{2}a(5)^2 \quad \rightarrow \quad a = 0.06 \text{ m/s}^2$$

b) Find angular acceleration $\alpha = \frac{a_t}{r} = \frac{0.06}{0.05} = 1.2 \text{ rad/s}^2$

c) Find torque $\tau_{net} = F_{t_{net}}\, r = (T_2 - T_1)\, R$

$$\tau_{net} = 0.33\ (0.05)$$

$$I = \frac{0.33(0.05)}{1.2} = 0.0138\ kg \cdot m$$

$$m_2 g - T_2 = m_2 a$$

$$T_1 - m_1 g = m_1 a$$

$$(T_1 - T_2) + (m_2 - m_1)g = +(m_2 + m_1)a$$

$$(T_2 - T_1) = 0.33\ N$$

Work in Rotational Motion

Change in system kinetic energy is done through work.

$$W = \Delta KE = KE_f - KE_i$$

$$= \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2$$

Work can also be defined in terms of "torque" and "displacement", as

$$W = \int_{\theta_i}^{\theta_f} \tau \, d\theta$$

For constant torque, it reduces to

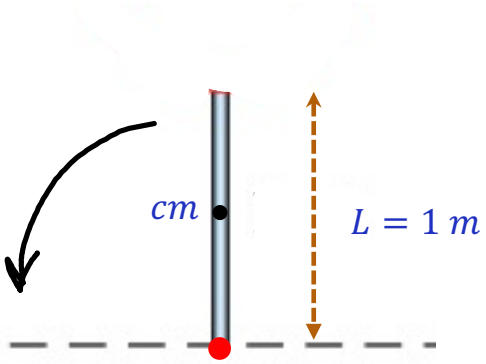
$$W = \tau \, (\theta_f - \theta_i)$$

We can write power as $$P = \frac{dW}{dt} = \tau \, \omega$$

Examples

1) Find the velocity of the tip of the rod as it hits the ground. Assume the center of mass at the mid-length of the rod. The rod is initially at rest.

The initial potential energy changes all to KE as the rod hits the ground.



No slipping

$$\frac{mgL}{2} \rightarrow KE$$

$$KE = \frac{1}{2} I \omega^2 = \frac{1}{2}\left(\frac{1}{12} mL^2 + m\left(\frac{L}{2}\right)^2\right)\omega^2 \quad = \frac{mgL}{2}$$

Solve for angular speed $\omega^2 = \frac{3g}{L}$

$$\left(\frac{v}{L}\right)^2 = \frac{3g}{L} \quad \rightarrow \quad v = \sqrt{3g} = 5.4\, m/s$$

Examples

2) The rotating wheel has to stop in 15 s. Find the necessary work to be done. The radius of the wheel is 1.2 meters and its mass is 32 kg.

$$KE = \frac{1}{2} I \omega^2 = \frac{1}{2}(mR^2)\omega^2$$

$$KE = \frac{1}{2}32(1.2)^2\left(\frac{280\ (2\pi)}{60}\right)^2 = 19.81\ kJ$$

$$W = -19.81\ kJ$$

$$P_{ave} = \frac{\Delta W}{\Delta t} = 1.33\ kW$$



$$\omega = 280\ rev/min$$

$$\omega = \frac{280\ (2\pi)}{60}\ rad/s$$

Chapter 11: Rolling and Angular Momentum

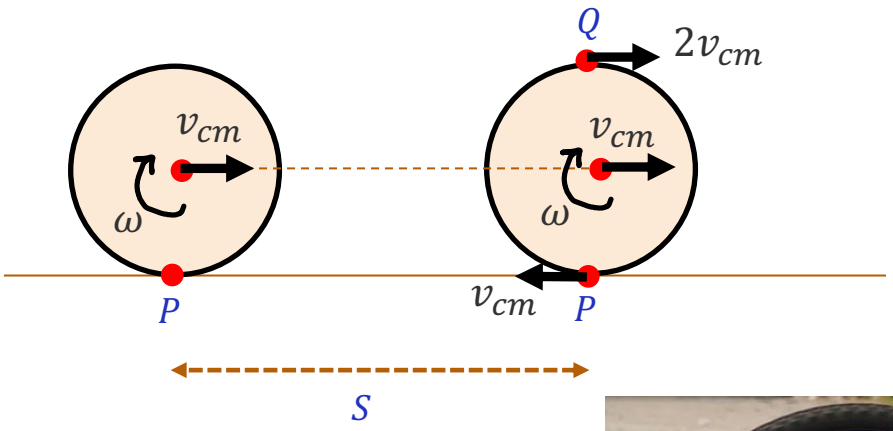
The following subjects will be discussed

- Rolling.
- Forces and Torque.
- Angular Momentum.
- Conservation of Angular Momentum

Rolling

Rolling is a combination of translational motion and rotational motion. We can think of it as a moving rotational axis (not fixed).

We will only consider "smooth" rolling.











In case of smooth rolling,

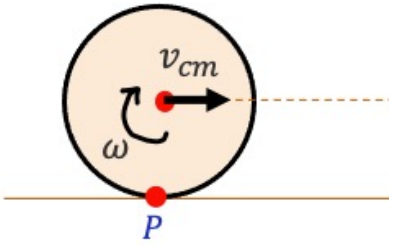
$$S = \theta R \qquad v_{cm} = \omega R$$

$$a_{cm} = \alpha R$$

$$KE = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} m v_{cm}^2$$

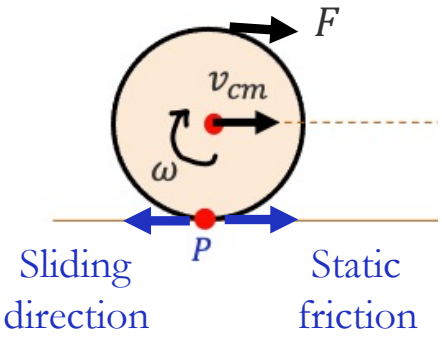
Forces in Rolling

1) For smooth rolling (no sliding) with constant speed, NO frictional forces are involved, even on rough surfaces. Remember that the point of contact P, has $v = 0$.

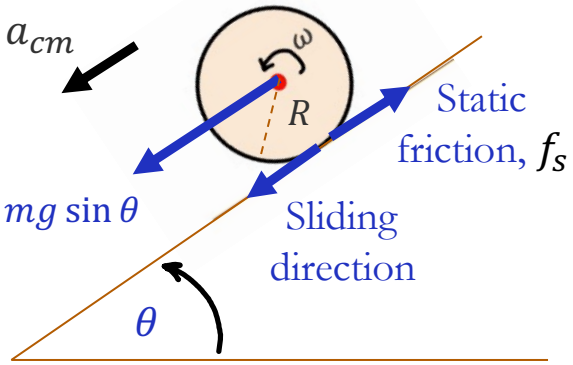


2) In case of acceleration (force that causes torque), the static friction will kick-in to resist the acceleration.

In case of sliding, the kinetic friction will be considered.



3) When external forces or gravity is involved.



For smooth rolling:

$$mg\sin\theta - f_s = m\,a_{cm} \quad \text{linear}$$

$$\tau_{net} = f_s \cdot R = I_{cm}\,\alpha = a_{cm}/R \quad \text{rotation}$$

$$\Rightarrow \quad a_{cm} = \frac{g\sin\theta}{1 + (I_{cm}/mR^2)}$$

Note: $f_s \neq \mu_s F_N$

General Expression for Torque

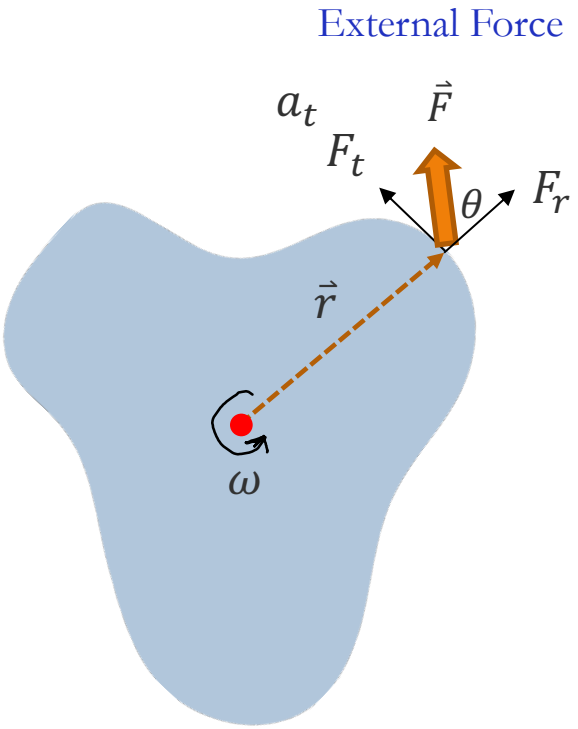
In 3D, the equation for torque need to be written in vector form.

$$\tau = F_t \; r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Position vector relative to axial point

Applied force



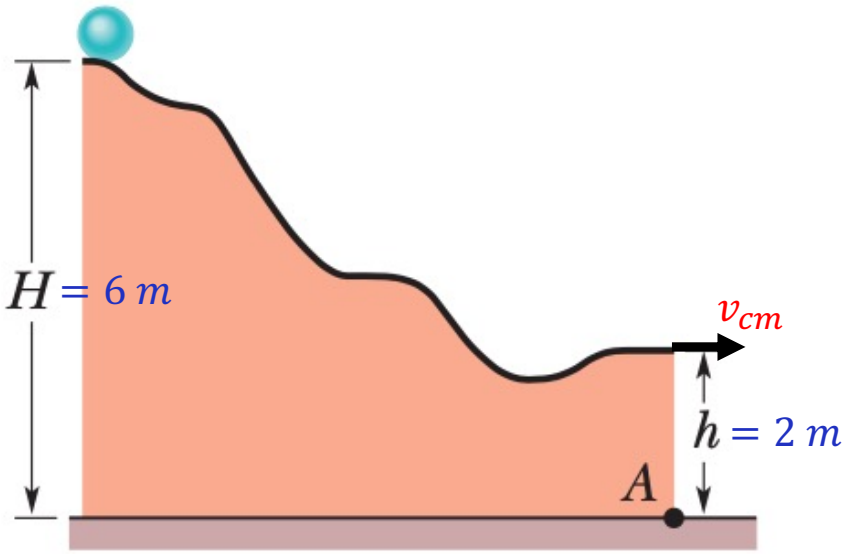
The resulting torque is in a direction normal to the plane of $\vec{r}$ and $\vec{F}$.

$$|\tau| = |\vec{r}||\vec{F}| \sin\theta$$

Examples

1) The ball (mass m) experiences smooth rolling throughout the trip. How far from point A will it land? The ball is initially at rest.

The initial potential energy changes all to KE as the ball reaches above A.



$$PE = mg(H - h)$$

$$KE = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} m v_{cm}^2$$

$$\frac{1}{2}\left(\frac{2}{5}mR^2\right)\frac{v_{cm}^2}{R^2} + \frac{1}{2} m v_{cm}^2 = mg(H - h)$$

$$v_{cm}^2 = \frac{20g}{3.5}$$

Projectile motion $\quad t = 0.64\ s \quad x = 4.8\ m$

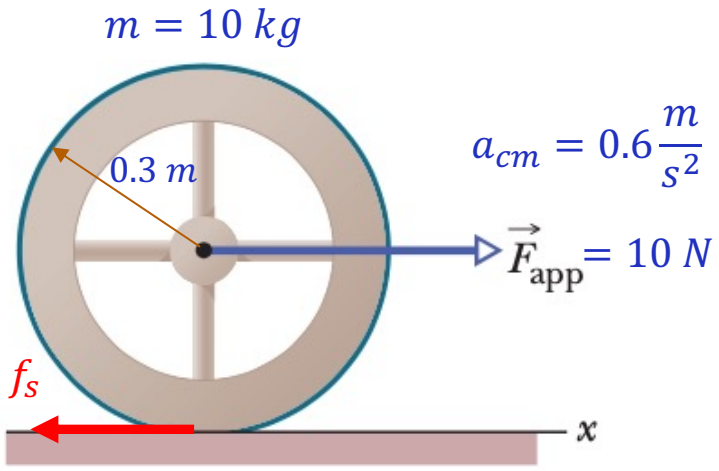
Examples

2) What frictional force necessary to keep an acceleration of the center of mass of the wheel at the given value?

$$F - f_s = m\, a_{cm}$$

$$f_s = F - m\, a_{cm} = 10 - (10)(0.6)$$

$$f_s = 4\ N$$



Find the inertia of the wheel.

$$\tau_{net} = f_s \cdot R = I_{cm}\, \alpha = I_{cm} \frac{a_{cm}}{R}$$

$$I_{cm} = \frac{f_s \cdot R^2}{a_{cm}} \quad \Rightarrow \quad I_{cm} = \frac{(4) \cdot (0.3)^2}{0.6} = 0.6 \frac{kg}{m^2}$$

Angular Momentum

Angular momentum is the momentum associated with angular motion. It is defined with respect to some axis of rotation.

For a single particle:

$$\vec{l} = \vec{r} \times \vec{p}$$

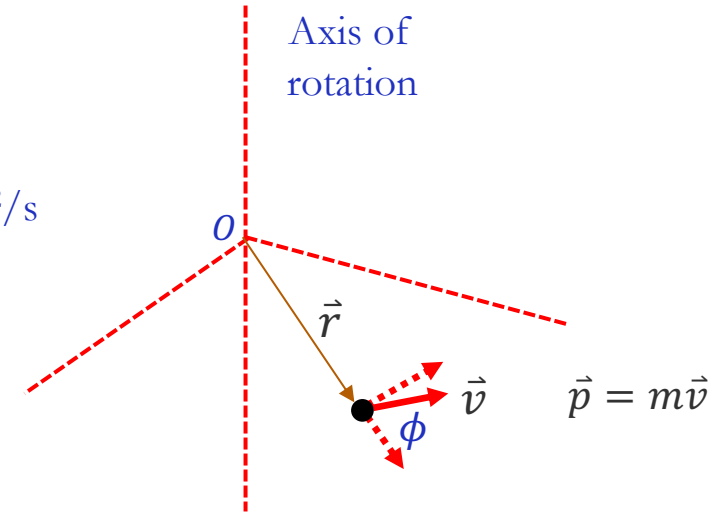
$$\vec{l} = m(\vec{r} \times \vec{v})$$

Kg m^2/s

Position vector relative to axial point

Linear momentum

$$|l| = m\,|\vec{r}||\vec{v}|\sin\phi$$



Angular momentum is in a direction normal to the plane of $\vec{r}$ and $\vec{v}$.

Counter-clockwise (CCW) rotation: Positive angular momentum.

Clockwise (CW) rotation: Negative angular momentum.

Note: Angular momentum is defined even with no rotation!!

$$\frac{d\vec{p}}{dt} = \vec{F}_{net}$$

$$\frac{d\vec{l}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F}_{net}$$

$$\frac{d\vec{l}}{dt} = \vec{\tau}_{net}$$

Angular Momentum

For Many particles:

$$\vec{L} = \vec{l}_1 + \vec{l}_2 + \cdots + \vec{l}_n = \sum_{i=1}^{n} \vec{l}_i$$

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^{n} \frac{d\vec{l}_i}{dt} = \sum_{i=1}^{n} \vec{\tau}_i = \vec{\tau}_{net}$$

Over the same axis of rotation

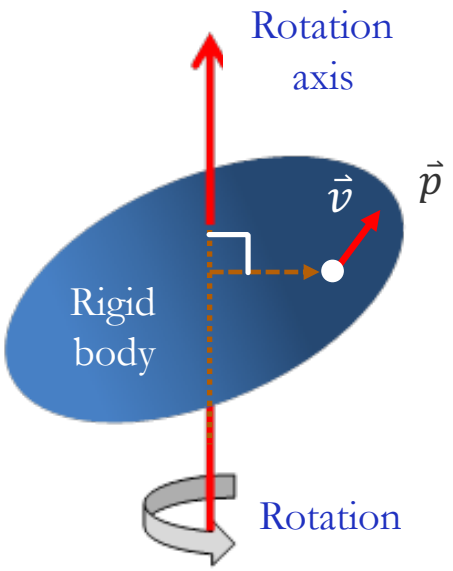
For a Rigid Body:

$$\vec{L} = \sum_{i=1}^{n} \vec{l}_i = \sum_{i=1}^{n} m_i \, (\vec{r}_i \times \vec{v}_i) = \sum_{i=1}^{n} m_i \, (\vec{r}_i \times \omega \vec{r}_i)$$

$$\vec{L} = \omega \sum_{i=1}^{n} m_i \, r_i^2 \qquad \vec{L} = I \, \omega$$

Over the same axis of rotation

$$\frac{d\vec{L}}{dt} = \vec{\tau}_{net}$$



Conservation of Angular Momentum

For a system with no net external torque ($\vec{\tau}_{net} = 0$), we have:

$$\frac{d\vec{L}}{dt} = 0 \quad \Rightarrow \quad \vec{L} = Constant$$

$$\vec{L}_{initial} = \vec{L}_{final}$$

Note that if $\tau_{net_x} = 0 \rightarrow L_x$ is conserved, and so on.

Interesting result

$$I_i\,\omega_i = I_f\,\omega_f$$

Change in shape (inertia)
causes
change in rotation speed.



Conservation of Angular Momentum



Q & A

Send me an email if
you have any
questions.